Semantic Similarity Index

ARI measuring the similarity of two semantic fields

Store a dataset of 1.8M word associations (e.g. “ship”→”sail”) in hetero-associative memory. Retrieve the activation map for multiple words. Measure their similarity via the Adjusted Rand Index.

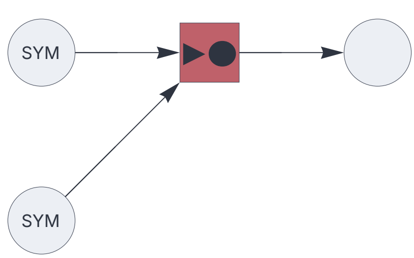
```
<< "Circuits.m"
```

Hetero-associative memory circuit

```
{n, p} = {5000, 10};
```

```
dataflow = {
        input[codec] [1] [],
        input[codec] [2] [],
        associator[] [11] [2, 1],
        output[] [] [11],
        _ → {n, p}
    };
```

```
c = Circuit[dataflow]; c[Graph, ImageSize→ 220]
```



Import dataset

```
data = Import[ NotebookDirectory[] <>  "/data/word-associations.csv.zip",
          "word-associations.csv"];
```

```
RandomSample[data, 10]
```

```
{dunes ⟷ habitat, upped ⟷ security, exercising ⟷ flexibility,
 segregate ⟷ distinguish, enthrallment ⟷ spell,
 ambergris ⟷ ocean, nature ⟷ beach, self-referential ⟷ self-aware,
 linearization ⟷ straight, intermix ⟷ splice}
```

```
Length[data]
```

```
1 888 505
```

Train

```
c @@@ data;
```

Adjusted Rand Index

```
ARI[A_List, B_List, dim_Integer] :=
N[(Length[Intersection[A,B]] - Length[A]*Length[B] / dim ) /
     (Min[Length[A], Length[B]] -  Length[A]*Length[B] / dim)]
```

```
similarity[x_, y_] := ARI[ c[x, Null], c[y, Null], n]
```

Test

Retrieve activation map.

```
c[ "cat", Null]
```

```
Sequence[{19, 23, 32, 43, 61, 66, 74, 80, 116, 126, 129, 145, 152, 188, 220,
  241, 296, 325, 342, 343, 370, 377, 385, 388, 419, 454, 456, 467, 495,
  500, 530, 534, 546, 548, 560, 584, 589, 610, 619, 633, 675, 687, 688,
  693, 697, 708, 710, 732, 754, 758, 773, 787, 794, 800, 803, 807, 837,
  864, 881, 925, 973, 986, 1015, 1037, 1040, 1045, 1049, 1052, 1064,
  1069, 1072, 1077, 1092, 1109, 1112, 1115, 1123, 1150, 1163, 1169, 1170,
  1188, 1196, 1200, 1216, 1219, 1230, 1277, 1295, 1302, 1345, 1376, 1384,
  1390, 1420, 1444, 1474, 1483, 1484, 1506, 1513, 1519, 1531, 1555, 1569,
  1574, 1582, 1592, 1593, 1598, 1617, 1640, 1648, 1672, 1692, 1723, 1731,
  1787, 1820, 1825, 1831, 1835, 1838, 1864, 1868, 1888, 1912, 1916, 1919,
  1922, 1935, 1950, 1958, 1983, 1996, 2008, 2017, 2018, 2049, 2070, 2075,
  2088, 2101, 2102, 2104, 2166, 2207, 2221, 2269, 2286, 2300, 2333, 2354,
  2395, 2402, 2406, 2426, 2434, 2438, 2463, 2502, 2503, 2521, 2559, 2575,
  2590, 2599, 2600, 2602, 2617, 2618, 2647, 2653, 2693, 2703, 2749, 2769,
  2790, 2798, 2809, 2817, 2825, 2828, 2830, 2839, 2858, 2962, 2967,
  2969, 2970, 3004, 3032, 3045, 3050, 3055, 3130, 3155, 3163, 3177,
  3187, 3199, 3207, 3248, 3250, 3260, 3263, 3275, 3277, 3327, 3329, 3330,
  3336, 3417, 3418, 3424, 3476, 3489, 3508, 3567, 3580, 3597, 3612,
  3632, 3635, 3648, 3674, 3681, 3702, 3717, 3735, 3736, 3742, 3756,
  3798, 3825, 3838, 3863, 3884, 3890, 3913, 3916, 3931, 3942, 3943, 3993,
  4010, 4014, 4015, 4050, 4114, 4116, 4168, 4211, 4232, 4242, 4250, 4259,
  4267, 4271, 4273, 4287, 4297, 4342, 4382, 4412, 4438, 4454, 4476, 4479,
  4481, 4499, 4520, 4523, 4531, 4565, 4571, 4582, 4584, 4662, 4703, 4709,
  4714, 4720, 4732, 4762, 4767, 4776, 4780, 4784, 4792, 4795, 4820, 4828,
  4831, 4856, 4858, 4866, 4868, 4894, 4917, 4928, 4956, 4975, 4982, 4991}]
```

Measure semantic similarity via ARI:

```
similarity["ship", "boat"]
```

```
0.614402
```

```
similarity["ship", "cheese"]
```

```
-0.0248165
```

Similarity Matrix

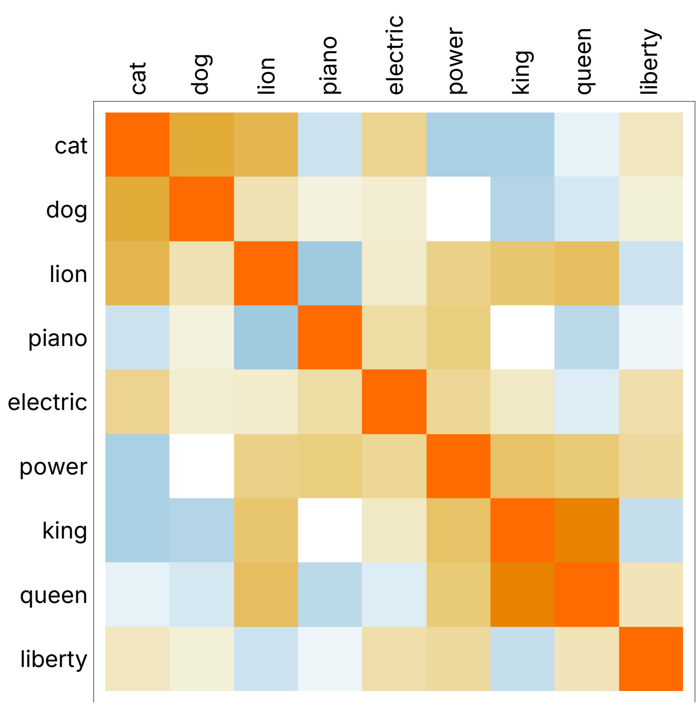
```
SimilarityMatrix[ items_List] := Module[ {simtable, xTicks, yTicks},

    simtable = Outer[ Round[similarity[##], 0.001]&, items, items];
    xTicks=Table[{i,Rotate[items[[i]],90 Degree], {0,0}},{i,Length[items]}];
    yTicks=Table[{i,items[[i]], {0,0}},{i,Length[items]}];

    MatrixPlot[simtable,
        FrameTicks → {{yTicks, None}, {None, xTicks}},
         FrameStyle → Opacity[0],
        FrameTicksStyle → Directive[Opacity[1],
        FontFamily → "Inter Variable",
        FontSize→12]]
    ]
```

```
items = {"cat",  "dog", "lion", "piano",
      "electric", "power", "king", "queen", "liberty"};
```

```
SimilarityMatrix[ items]
```



```
Export["similaritymatrix.png", ImagePad[Rasterize[%], 30, White]];
```